

How Did COVID-19 School Closures Amplify Human-Capital Inequality Across Socioeconomic Groups in India? Evidence from Learning Outcomes, Remote Learning Access, and Access to Key Learning Inputs

Description, Motivation and Background:

Human capital¹ is a key input into labour productivity and an “important determinant of growth” in growth theory.² Therefore, school closures matter economically because they are not just a short-run disruption to learning, but a negative shock to human-capital accumulation. Cohen et al. (2022) quantify the potential economic impacts of COVID-19-related school closures, estimating that secondary-school closures in India will lead to an annual average shock to labour productivity of 2.145% once affected cohorts have entered the labour force.³ This productivity shock is projected to contribute to earning losses of 4.70%,⁴ making it harder for some individuals to meet their basic needs.⁵ In macroeconomic terms, these earning losses in India are estimated to lead to a GDP shortfall of \$98.84 billion in 2030, the greatest absolute GDP shortfall in South Asia.⁶ This suggests that human-capital losses due to school closures may create long-run development costs for India by weakening future labour productivity and reducing GDP.

These high projected costs are consistent with the scale of school disruption in India; schools were closed for 82/104 weeks between March 2020 and March 2022, making India’s closures the “joint-second longest duration of any country.”⁷ Figure 1 shows a drastic fall in daily learning time,⁸ contributing to severe learning losses during COVID for Indian children. This is reflected in ASER evidence on foundational reading; the proportion of Grade 3

¹ **Human Capital** – The accumulated education, health, and skills (e.g., literacy/numeracy, non-cognitive, and digital skills) of individuals that enable productivity in the labour market.

² De La Maisonnette, C., Égert, B., & Turner, D. (2022). Quantifying the macroeconomic impact of COVID-19-related school closures through the human capital channel. *OECD*. Page 7, <https://doi.org/10.1787/eea048c5-en>

³ Cohen, S., Chakravarthy, S., Bharathi, S., Narayanan, B., & Park, C. (2022). Potential economic impact of COVID-19-Related school closures. *Asian Development Bank*. Page 7, <https://doi.org/10.22617/wps220197-2>

⁴ Cohen, S., Chakravarthy, S., Bharathi, S., Narayanan, B., & Park, C. (2022). Potential economic impact of COVID-19-Related school closures. *Asian Development Bank*. Page 7, <https://doi.org/10.22617/wps220197-2>

⁵ Cohen, S., Chakravarthy, S., Bharathi, S., Narayanan, B., & Park, C. (2022). Potential economic impact of COVID-19-Related school closures. *Asian Development Bank*. Page 3, <https://doi.org/10.22617/wps220197-2>

⁶ Cohen, S., Chakravarthy, S., Bharathi, S., Narayanan, B., & Park, C. (2022). Potential economic impact of COVID-19-Related school closures. *Asian Development Bank*. Page 10, <https://doi.org/10.22617/wps220197-2>

⁷ Andrew, A., & Salisbury, A. (2023). The educational experiences of Indian children during COVID-19. *Economics of Education Review*, 97, 102478. Page 3, <https://doi.org/10.1016/j.econedurev.2023.102478>

⁸ Andrew, A., & Salisbury, A. (2023). The educational experiences of Indian children during COVID-19. *Economics of Education Review*, 97, 102478. Page 4, <https://doi.org/10.1016/j.econedurev.2023.102478>

students able to read a Grade 2-level text fell from 27.3% in 2018 to 20.5% in 2022.⁹ There was also widespread regression in children’s learning,¹⁰ with 82% of children losing at least one specific mathematical ability.¹¹ Given the scale of this learning shock, it is important to question whether human-capital losses were larger for more **socioeconomically disadvantaged children**¹², as these children were more likely to face barriers to remote learning and other key learning inputs.

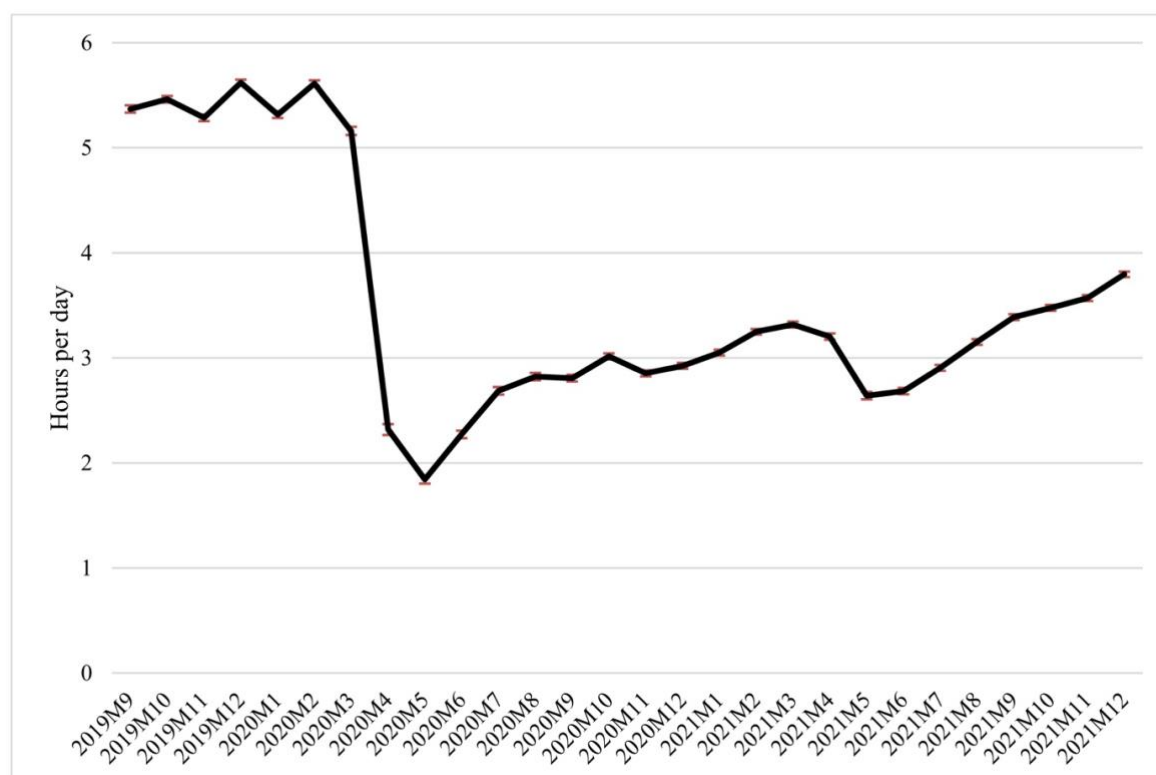


Figure 1: Plots month-by-month average daily learning time for 12-18-year-olds across all Indian states, Sep 2019 – Dec 2021⁸

⁹ ASER Centre. (2019). *Enrollment and learning report card: From the annual status of education report 2018*. Page 2, <https://img.asercentre.org/docs/ASER%202018/Release%20Material/enrollmentandlearningenglish.pdf>

ASER Centre. (2023). *Enrollment and learning report card: From the annual status of education report 2022*. Page 2, https://img.asercentre.org/docs/ASER%202022%20report%20pdfs/All%20India%20documents/Enrollment%20and%20Learning%20Report%20Card/Enrollment%202022_English.pdf

¹⁰ Azim Premji Foundation. (2021). *Loss of learning during the pandemic*. Page 3, <https://azimpremjiuniversity.edu.in/media/publications/downloads/report/Learning-Loss-during-pandemicc.f1712062129.pdf>

¹¹ Azim Premji Foundation. (2021). *Loss of learning during the pandemic*. Page 10, <https://azimpremjiuniversity.edu.in/media/publications/downloads/report/Learning-Loss-during-pandemicc.f1712062129.pdf>

¹² Socioeconomically disadvantaged children – Refers to children (aged 0-17) from households with lower parental education, lower household wealth, and weaker access to digital resources (e.g., computers, smartphones, and internet access).

This question is important because even before COVID-19, India had substantial **human-capital inequality**¹³ by **socioeconomic status (SES)**¹⁴. As Figure 2 shows, literacy rates differ by household wealth; between 2009–2014, around 80% of 16 year olds in the richest 20% of households were literate, compared with only 50% in the poorest 40% of households.¹⁵ Figure 3 similarly highlights learning gaps linked to SES; SES is the strongest predictor of subtraction ability in primary schools, and by Grade 5 children from the poorest households were 19 percentage points less likely to subtract correctly than those from the wealthiest households.¹⁶ These gaps matter because literacy and numeracy are foundational skills for later learning and work,¹⁷ and lacking these skills can limit “prospects for decent work”¹⁸ and reduce long-run earnings opportunities. So, if COVID-19 school closures generated larger human-capital losses for socioeconomically disadvantaged children, then these pre-existing human-capital inequalities will have widened even further, causing greater inequality of opportunity in the labour market.

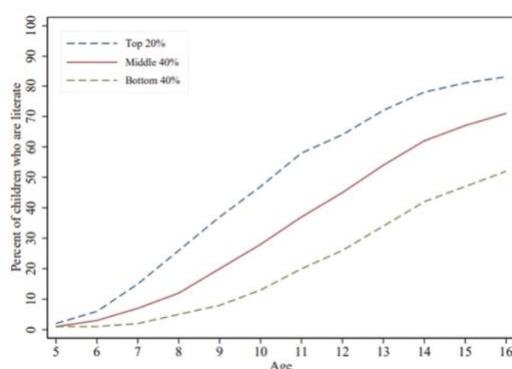


Figure 2: Plots literacy rate by age and household wealth ¹⁵

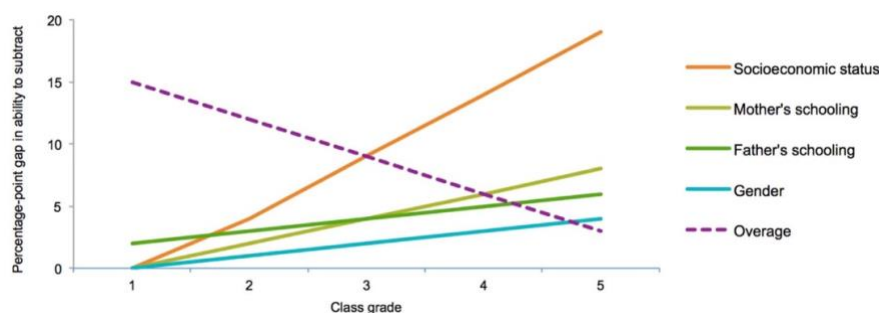


Figure 3: Shows percentage-point gaps in subtraction ability across student background factors from Grades 1—5 ¹⁶

¹³ Human-Capital Inequality – Differences in human-capital stocks (e.g., education, health, and skills) between individuals or groups that translate into unequal productivity and long-run earnings opportunities.

¹⁴ Socioeconomic Status (SES) – An individual’s or household’s relative position in the social and economic hierarchy, captured by parental education, household wealth, parental occupations, and access to resources (e.g., digital).

¹⁵ Akmal, M., & Pritchett, L. (2021). Learning equity requires more than equality: Learning goals and achievement gaps between the rich and the poor in five developing countries. *International Journal of Educational Development*, 82, 102350. Page 7, <https://doi.org/10.1016/j.ijedudev.2021.102350>

¹⁶ Alcott, B., & Rose, P. (2017). Learning in India’s primary schools: How do disparities widen across the grades? *International Journal of Educational Development*, 56, 42—51. Page 48, <http://dx.doi.org/10.1016/j.ijedudev.2017.05.002>

¹⁷ World Bank. (2025, May 16). *Foundational Learning*. <https://www.worldbank.org/en/topic/education/brief/foundational-learning>

¹⁸ UNESCO. (2025, June 2). *Foundational Learning*. <https://www.unesco.org/en/quality-learning/foundational>

It is plausible that school closures widened these pre-existing skill gaps across socioeconomic groups because of the shift to remote and home-based learning. Many schools “struggled to provide virtual alternatives to in-person teaching,”¹⁹ and many families “lacked the technology, space and equipment” required for effective learning.²⁰ These constraints were strongly linked to SES; higher-income and more educated households had greater “access to big-screen devices” and remote education,²¹ as well as key learning inputs, including private tuition.²² Furthermore, in 2018, only 2.7% of the poorest 20% of households had access to a computer and 8.9% had internet access, compared with 27.6% and 50.5% amongst the richest 20% of households.²³ This digital divide likely widened pre-existing learning gaps by limiting poorer children’s ability to benefit from remote learning. Beyond digital access, household education is another plausible socioeconomic driver of unequal learning losses during COVID-19, because learning support at home is an important determinant of academic achievement.²⁴

Therefore, my analysis tests the hypothesis that **human-capital losses, particularly learning losses, during COVID-19 were larger for children from more ‘disadvantaged socioeconomic backgrounds’¹².**

¹⁹ Andrew, A., & Salisbury, A. (2023). The educational experiences of Indian children during COVID-19. *Economics of Education Review*, 97, 102478. Page 1, <https://doi.org/10.1016/j.econedurev.2023.102478>

²⁰ Andrew, A., & Salisbury, A. (2023). The educational experiences of Indian children during COVID-19. *Economics of Education Review*, 97, 102478. Page 1, <https://doi.org/10.1016/j.econedurev.2023.102478>

²¹ LIRNEasia, & ICRIER. (2021, November 12). *Digital connectivity more than doubles during COVID shutdowns in India. But remote work and remote education gains unimpressive*. LIRNEasia. Page 1, <https://lirneasia.net/wp-content/uploads/2021/11/Digital-connectivity-more-than-doubles-during-COVID-shutdowns-in-India.-But-remote-work-and-remote-education-gains-unimpressive.pdf>

²² Vyas, A. (2025). Parental involvement in children’s learning in India during the COVID-19 pandemic. *Hillary Place Papers*, 9, 6—12. Page 9, <https://doi.org/10.48785/100/335>

²³ Mahendru, A., Dutta, M., & Mishra, P. R. (2022). *Digital divide: India inequality report 2022*. OXFAM India. Page 15, <https://ruralindiaonline.org/en/library/resource/digital-divide-india-inequality-report-2022/>

²⁴ Banerji, R., & Wadhwa, W. (2023). COVID-19 and global education: Evidence from India. In C. Chatterjee, A. S. Chakrabarti, & A. B. Deolalikar (Eds.), *Flattening the Curve: COVID-19 & grand challenges for global health, innovation, and economy* (pp. 383—419). World Scientific Publishing Company. Page 395, https://asercentre.org/wp-content/uploads/2022/12/covid-19-and-global-education_RB_WW.pdf

Analysis:

To test my hypothesis, I assess whether learning losses during COVID-19 school closures in India differed by three socioeconomic indicators: household education, household wealth, and digital access. These indicators capture differences in a household's ability to support effective learning at home, so I predict larger losses amongst children from more disadvantaged socioeconomic backgrounds. I use evidence from learning outcomes and access to remote learning and other key learning inputs to examine whether human-capital losses were indeed regressive across socioeconomic groups.

Using data on primary school children in Tamil Nadu²⁵, Singh et al. (2022) quantify learning losses during COVID-19 and show that they vary by maternal education. Table 1 reports learning loss in standard-deviation (SD) units by comparing children's December 2021 and August 2019 test scores, controlling for village fixed effects, age, and gender. For children whose mothers had primary education or less, scores in December 2021 were 0.750 SD lower in maths and 0.370 SD lower in Tamil than those of the same age in the same villages in August 2019.²⁶ However, children whose mothers completed Grade 12+ experienced smaller learning losses in maths by 0.100 SD, and in Tamil by 0.071 SD, with these differences being statistically significant at the 5% and 1% levels respectively.²⁷ This is consistent with the view that more educated households were better able to protect children's learning during closures by providing greater learning support.²⁸

Maternal Education Group	Math Learning Loss	Tamil Learning Loss
Primary or Less	-0.750	-0.370
Incomplete Secondary	-0.727	-0.360
Grade 12 +	-0.650	-0.299

Table 1: Shows estimated learning losses (in standard-deviation units) in maths and Tamil between Aug 2019 and Dec 2021, by maternal education group ²⁶

²⁵ Tamil Nadu – Located in South India, and Tamil is its official language.

²⁶ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 21, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

²⁷ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 21, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

²⁸ Vyas, A. (2025). Parental involvement in children's learning in India during the COVID-19 pandemic. *Hillary Place Papers*, 9, 6—12. Page 11, <https://doi.org/10.48785/100/335>

Singh et al. (2022) argue that mothers' education was "significantly [positively] correlated with student access to educational inputs" and learning resources during school closures.²⁹ This is illustrated in Figure 4. For example, access to video classes rises from 8% to 21% as maternal education increases, and the share of children receiving homework from school rises from 13% to 26%.³⁰ Importantly, Singh et al. (2022) show that these inputs are associated with learning progress by estimating their "value added" on maths and Tamil test scores.³¹ Using regression estimates, they suggest that access to video classes is linked to +0.20 maths value added and +0.081 Tamil value added, whilst receiving homework from school is linked to +0.15 in maths and +0.046 in Tamil.³² These estimates are all statistically significant at the 1% level ($p < 0.01$).

This evidence suggests that school closures likely amplified human-capital inequality across socioeconomic groups in India, because children from less educated households had less access to learning inputs that significantly improved achievement, and therefore they experienced larger learning losses, reflected in greater declines in test scores.

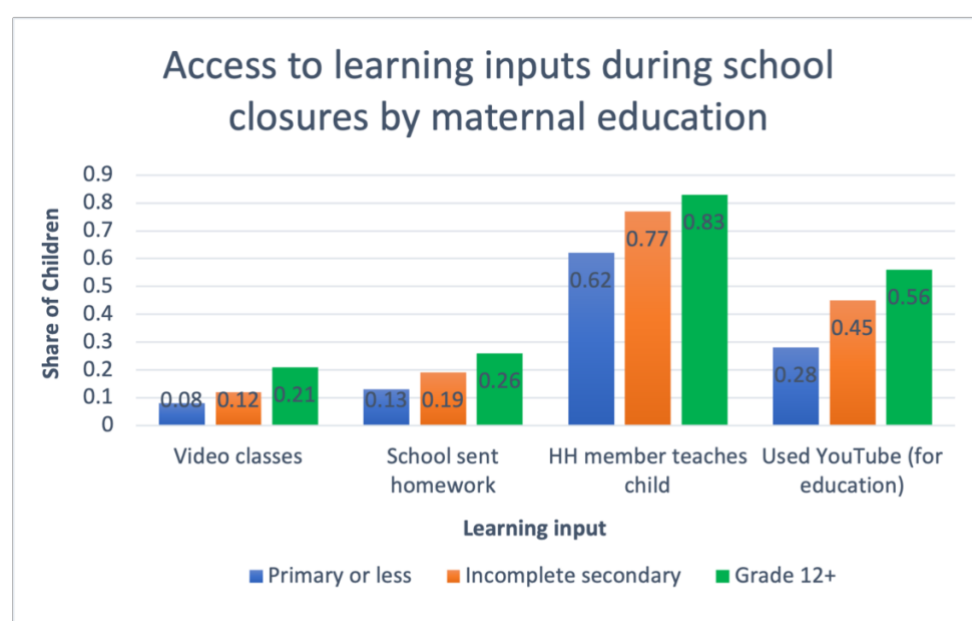


Figure 4: Mapping children's access to key learning inputs by maternal education group in Tamil Nadu during school closures³⁰

²⁹ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 8, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

³⁰ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 22, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

³¹ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 22, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

³² Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 22, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

Exploring the impacts of school closures on learning losses by household wealth further supports my hypothesis. Using data on primary school children in Assam³³, Guariso & Nyqvist (2023) compare test scores from 2019 and 2022. Table 2 shows that language learning losses were 0.224 SD larger for children from below-median wealth households compared with those from above-median wealth households ($p < 0.01$).³⁴ However, maths learning losses were statistically insignificant.

Household Wealth Group	Math Learning Loss	Language Learning Loss
Below-median	-0.245	-0.406
Above-median	-0.171	-0.182

Table 2: Shows estimated learning losses (in standard-deviation units) in maths and language between 2019 and 2022, by household wealth group³⁴

National evidence from Nandi et al. (2023) compares changes in literacy for 15-17-year-old Indian school children before and after the school closures across household wealth quintiles. Using a matching³⁵ method, they estimate that girls in the poorest wealth quintile experienced the greatest shock in literacy, with Table 3 showing a 3.1-3.8% reduction in reading ability ($p < 0.01$).³⁶ The shock for girls in wealth quintiles 2-4 was 1.0-2.1%, but, interestingly there is no evidence of literacy decline for girls from the richest wealth quintile.³⁷ For boys, they find no statistically significant evidence of reduction in literacy due to the pandemic. This difference in impact between girls and boys may be due to the unequal sample sizes, or because girls faced more barriers to studying at home; 67% of young women reported spending more time on household chores during school closures than before COVID-19, compared to 37% of young men.³⁸

Taken together, both studies provide evidence that learning losses in India were greater for children from poorer households, supporting the hypothesis that school closures had regressive impacts and widened human-capital inequality by household wealth.

³³ Assam – A state in the North-East of India.

³⁴ Guariso, A., & Nyqvist, M. B. (2023). The impact of the COVID-19 pandemic on children's learning and wellbeing: Evidence from India. *Journal of Development Economics*, 164, 103133. Page 6, <https://doi.org/10.1016/j.jdeveco.2023.103133>

³⁵ Matching – Nandi et al. (2023, page 2) match each post-COVID adolescent to the most similar pre-COVID adolescent(s) in the sample (using a propensity score). Then, they compare reading ability between the matched groups.

³⁶ Nandi, A., Haberland, N., Kozak, M., & Ngô, T. D. (2023). The gendered effects of the COVID-19 pandemic on adolescent literacy and schooling outcomes in India. *npj Science of Learning*, 8, Article 42. Page 4, <https://doi.org/10.1038/s41539-023-00193-8>

³⁷ Nandi, A., Haberland, N., Kozak, M., & Ngô, T. D. (2023). The gendered effects of the COVID-19 pandemic on adolescent literacy and schooling outcomes in India. *npj Science of Learning*, 8, Article 42. Page 4, <https://doi.org/10.1038/s41539-023-00193-8>

³⁸ Ford, K., & Singh, R. (2021). *Supporting vulnerable girls and young women in India: Evidence from the Listening to Young Lives at Work COVID-19 phone survey*. Young Lives. Page 4, <https://www.younglives.org.uk/sites/www.younglives.org.uk/files/YL-PolicyBrief-47-Apr21-4.pdf>

Wealth quintile	Girls: nearest neighbour	Girls: three nearest neighbours	Girls: sample size	Boys: nearest neighbour	Boys: three nearest neighbours	Boys: sample size
1	-3.1%	-3.8%	23,422	-2.4%	-3.0%	2938
2	-1.5%	-1.0%	23,724	0.1%	0.8%	3139
3	-2.1%	-1.6%	19,831	1.5%	0.7%	2698
4	-2.1%	-1.8%	16,160	-3.7%	-2.8%	2218
5	0.0%	0.3%	11,731	0.6%	0.7%	1780

Table 3: Shows changes in adolescents' reading ability before vs after COVID-19, reported across household wealth quintiles (wealth quintile 1 is the poorest) ³⁶

Figure 5 highlights a clear positive correlation between higher SES and access to 'high-tech learning modalities' and digital resources during school closures; the share of children receiving video classes rises from 7% in the bottom SES tercile to 22% in the top tercile ($p < 0.01$), and the share using YouTube for educational content increases from 31% to 56% ($p < 0.01$).³⁹ Van Cappelle et al. (2021) define 'high-tech learning modalities' as live video classes, YouTube videos, WhatsApp and learning apps, noting that they require access to a smartphone or computer.⁴⁰ Importantly, using a UNICEF survey across six Indian states, they find that use of these 'high-tech learning modalities' is associated with greater perceived learning for adolescents; using their estimates, I calculate that on average, using 'high-tech learning modalities' corresponds to around 1.59x higher odds of reporting higher levels of perceived learning during closures.^{41 42} Although they measure perceived learning rather than learning losses through test-score comparisons, their findings still support my hypothesis; relative to higher-SES children, lower-SES children had less access to digital resources and 'high-tech learning modalities,' both of which are linked to stronger perceived learning during closures, making it likely that school closures widened the pre-existing learning gaps in India and amplified human-capital inequality across socioeconomic groups.

³⁹ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 31, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

⁴⁰ van Cappelle, F., Chopra, V., Ackers, J., & Gochyyev, P. (2021). An analysis of the reach and effectiveness of distance learning in India during school closures due to COVID-19. *International Journal of Educational Development*, 85, 102439. Page 4, <https://doi.org/10.1016/j.ijedudev.2021.102439>

⁴¹ van Cappelle, F., Chopra, V., Ackers, J., & Gochyyev, P. (2021). An analysis of the reach and effectiveness of distance learning in India during school closures due to COVID-19. *International Journal of Educational Development*, 85, 102439. Page 6, <https://doi.org/10.1016/j.ijedudev.2021.102439>

⁴² Method used to calculate the average – From figures in Table 2 of van Cappelle et al. (2021, p.6-7):

For 'live video lessons' – $e^{0.604} = 1.8294 = 1.83$ ($p < 0.01$)

For 'WhatsApp, YouTube, learning applications on a smartphone' – $e^{0.294} = 1.3418 = 1.34$ ($p < 0.01$)

Then, I calculated the mean – $(1.8294 + 1.3418) \div 2 = 1.59$

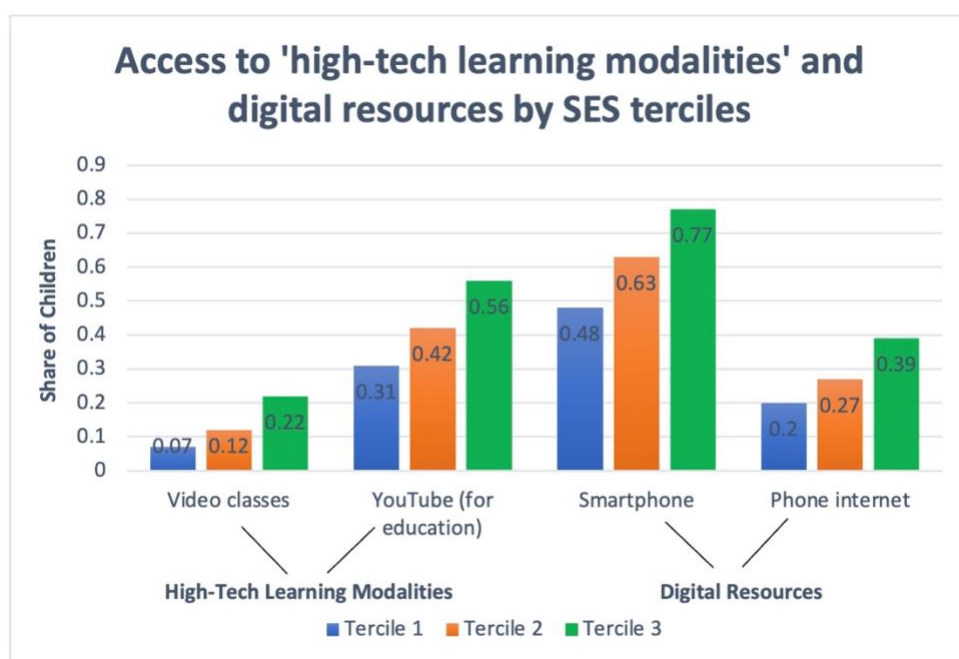


Figure 5: Mapping children's access to 'high-tech learning modalities' and digital resources during school closures, by SES tercile ³⁹

Conclusion & Next Steps:

The analysis explores the regressive tendencies of pandemics and provides convincing evidence that severe school closures in India amplified human-capital inequality across socioeconomic groups. Across the test-score and survey evidence, learning losses were disproportionately larger for children from more disadvantaged socioeconomic backgrounds, particularly those from less educated and poorer households. Crucially, whilst the magnitudes of learning losses vary across studies by state, subject, and gender, the overall outcomes are consistently regressive across socioeconomic groups.

One important takeaway from the analysis is that COVID-19 widened an already large learning gap by shifting learning into the home, where children's ability to maintain learning "heavily depended on the resources and support available at home."⁴³ Access to effective learning inputs and 'high-tech learning modalities' had clear "socioeconomic gradients"⁴⁴ (by maternal education and SES tercile).⁴⁵ Combining this with evidence that these inputs

⁴³ Guariso, A., & Nyqvist, M. B. (2023). The impact of the COVID-19 pandemic on children's learning and wellbeing: Evidence from India. *Journal of Development Economics*, 164, 103133. Page 6, <https://doi.org/10.1016/j.jdeveco.2023.103133>

⁴⁴ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 2, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

⁴⁵ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 22 & Page 31, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

are associated with value-added to test scores⁴⁶ as well as higher perceived learning⁴⁷ helps explain why learning losses were larger amongst disadvantaged children. Although we can infer that access to digital resources and key learning inputs benefits learning, some of the evidence relies on survey-based perceived learning. Therefore, this conclusion should be interpreted more cautiously than the household wealth and household education conclusions, which are based on measured learning outcomes (test-score and literacy evidence).

My findings suggest that post-pandemic learning recovery in India should have prioritised children from more disadvantaged backgrounds who experienced the largest setbacks, especially in foundational literacy and numeracy. Once schools reopened, targeted catch-up learning for low-performing students (e.g., additional teaching time) could have helped close the learning gaps widened by COVID-19 and reduced the risk of persistent human-capital inequality.

National evidence suggests larger literacy setbacks for girls in poorer households,⁴⁸ which may reflect gender norms that reduced girls' focus on education during school closures.⁴⁹ This motivates further research into the extent to which COVID-19 widened the gender learning gap. Further research should also examine other components of human capital, particularly health; disruptions to the **Midday Meal Scheme**⁵⁰ affected poorer children the most and contributed to much higher anaemia rates during closures.⁵¹ Therefore, it is plausible that this nutrition shock disproportionately affected disadvantaged children's health and ability to learn, and may have amplified human-capital inequality further across socioeconomic groups.

Word Count: 2061

⁴⁶ Singh, A., Romero, M., & Muralidharan, K. (2022). *COVID-19 learning loss and recovery: Panel data evidence from India* (RISE Working Paper 22/112). RISE. Page 22 & Page 31, https://riseprogramme.org/sites/default/files/2022-09/COVID-19_Learning_Loss_Recovery_Panel_Data_Evidence_India.pdf

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⁴⁸ Nandi, A., Haberland, N., Kozak, M., & Ngô, T. D. (2023). The gendered effects of the COVID-19 pandemic on adolescent literacy and schooling outcomes in India. *npj Science of Learning*, 8, Article 42. Page 4, <https://doi.org/10.1038/s41539-023-00193-8>

⁴⁹ UNICEF ROSA. (2020). *Gender-responsive education in the context of COVID-19: Framework and progressive standards for South Asia*. UNICEF. Page 8, <https://www.unicef.org/rosa/media/12361/file/Gender-responsive%20Education%20in%20the%20Context%20of%20Covid-19.pdf>

⁵⁰ **Midday Meal Scheme – A government programme in India that provides a free school lunch to children in government and government-aided schools.**

⁵¹ Thankachan, P., Selvam, S., Narendra, A. R., Mishra, H. N., Sachdev, H. S., Thomas, T., & Kurpad, A. V. (2022, July 5). There should always be a free lunch: The impact of COVID-19 lockdown suspension of the mid-day meal on nutriture of primary school children in Karnataka, India. *BMJ Nutrition, Prevention & Health*, 5(2), 364–366. Page 364, <https://nutrition.bmj.com/content/bmjnp/5/2/364.full.pdf>

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